

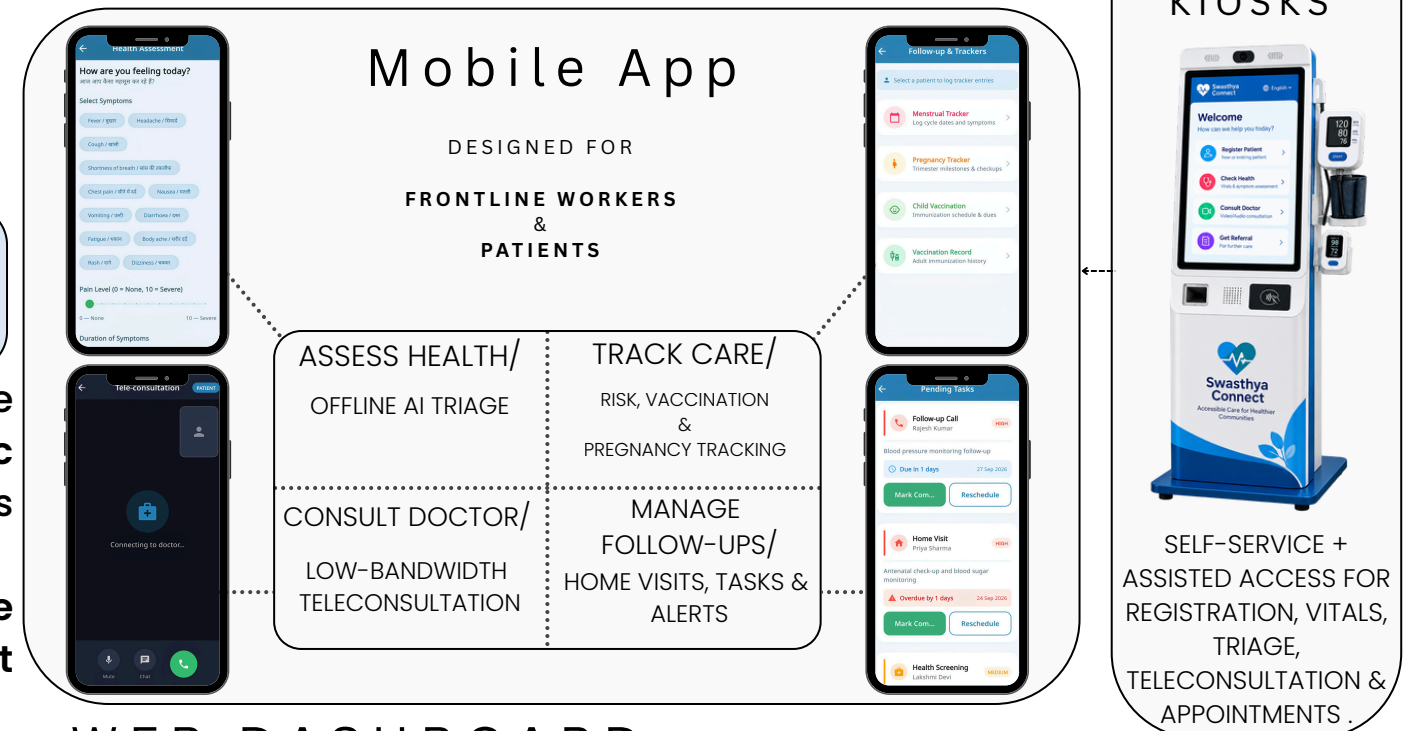
- **Problem Statement ID – SIH26133**
- **Problem Statement Title – Accessibility And Quality Of Public Healthcare Services, Particularly In Rural And Underserved Areas.**
- **Theme – MedTech / BioTech / HealthTech**
- **PS Category – Software**
- **Team ID – 165109**
- **Team Name (Registered on portal) – HealthSync1**



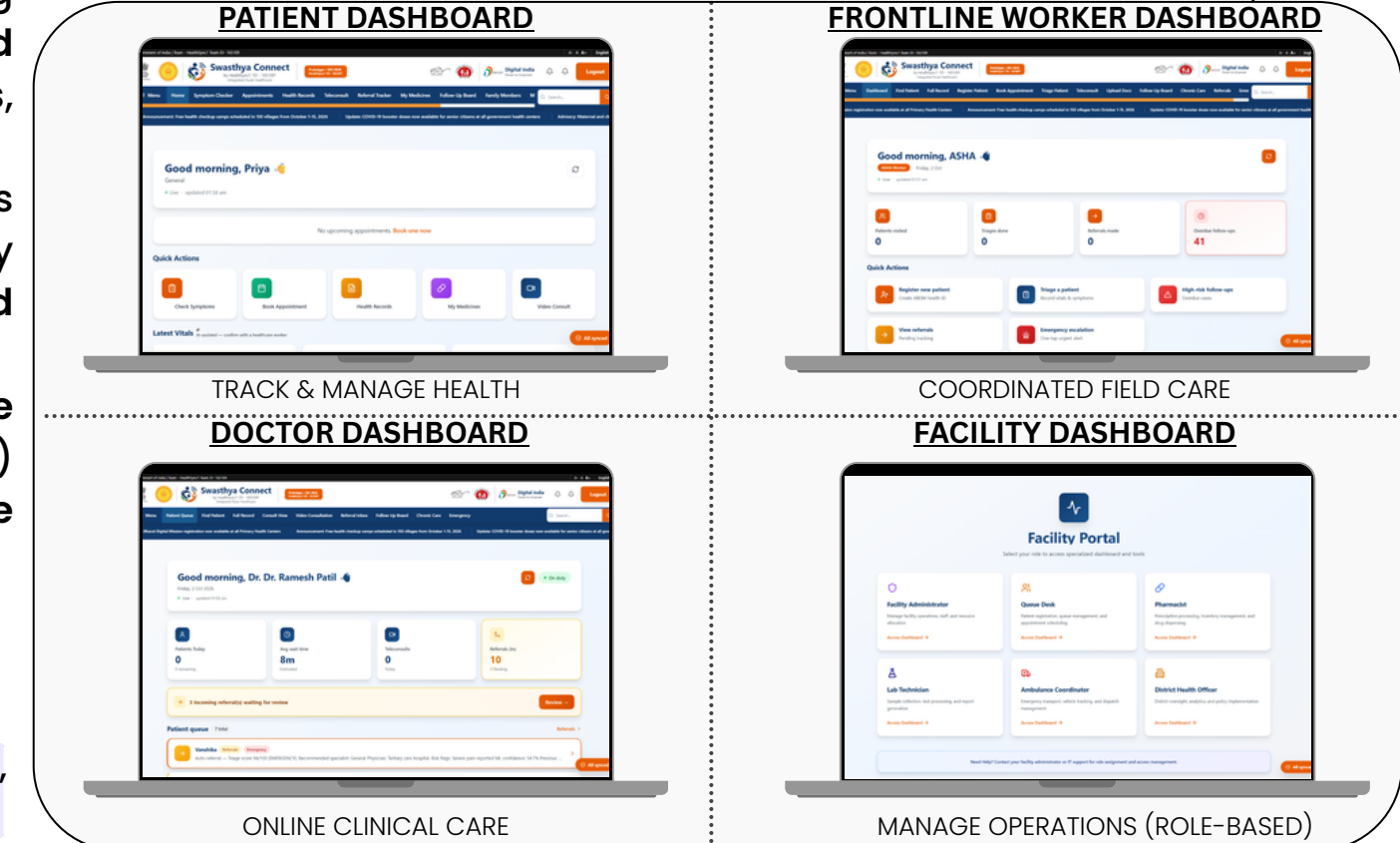
SOLUTION

Swasthya Connect is a healthcare coordination platform integrating Mobile App, Assisted Health Kiosks, Web Dashboards and IoT for accessible, connected care in rural and underserved communities.

PROTOTYPE



WEB DASHBOARD



Maternal, Child And Chronic Care
Dedicated workflows for maternal, child and chronic care enable automated alerts, scheduled care and frontline-worker follow-up.

MOBILE APP

PATIENT APP FOR CARE FROM HOME + OFFLINE FIRST FRONTLINE WORKER APP FOR FIELD VISITS.

ASSISTED HEALTH KIOSKS

SELF-SERVICE + ASSISTED CARE FOR PATIENT REGISTRATION, VITALS, DIGITAL TRIAGE & TELECONSULTATION

WEB DASHBOARD

ROLE-BASED DASHBOARDS FOR PATIENTS, FRONTLINE WORKERS, DOCTORS & FACILITIES.

- **Assisted Teleconsultation** : Video consultation with automatic audio fallback, AI-generated patient history summaries, and a female doctor preference(women patients) option. Urgency-based auto-booking secures the earliest suitable slot (e.g., 9:00–9:15 AM) within <5 minutes and instantly notifies the doctor.
- **Digital Triage** : Uses conversational AI-led assessment, symptoms and vitals (IoT where available) with a lightweight offline model to classify Routine, Priority or Critical and guide patients to the appropriate level of care.
- **Intelligent Longitudinal Patient Record** : OCR extracts medical records to create a chronological health timeline with AI summaries, alerts, and consent-based sharing.
- **Emergency Escalation** : High-risk triage/referral/patient SOS triggers the system to identify a suitable nearby facility using live location, and coordinate frontline worker/caregiver, doctor, and ambulance support.
- **Smart Referral Tracking** : Uses a unique referral ID to route and track patients across facilities, with automatic notifications, progress tracking and follow-ups until care is completed, ensuring continuity of care & accountability.
- **Yojna Engine & MediBot** : Yojna Engine recommends scheme with eligibility, while MediBot provides a multilingual chat assistant for patients and frontline workers.
- **High-Risk Follow-Up** : Provides timeline-based care tracking for maternal, child and elderly patients, with automated alerts for care needs, scheduled checkups and follow-ups, ensuring continuity of care.
- **Diagnostic & Medicine Coordination** : Connects prescriptions, diagnostic requirements and facility availability to help patients find and access required medicines and diagnostic services.
- **Multilingual Voice & Multimodal Assessment** : Voice (AI4Bharat ASR) + Text + Image (wounds/infections) capture symptoms, severity & duration, followed by adaptive questioning.

Our solution digitizes healthcare workflows, enabling continuous patient monitoring, real-time updates, and coordinated care across facilities.

WHY WE STAND OUT ?

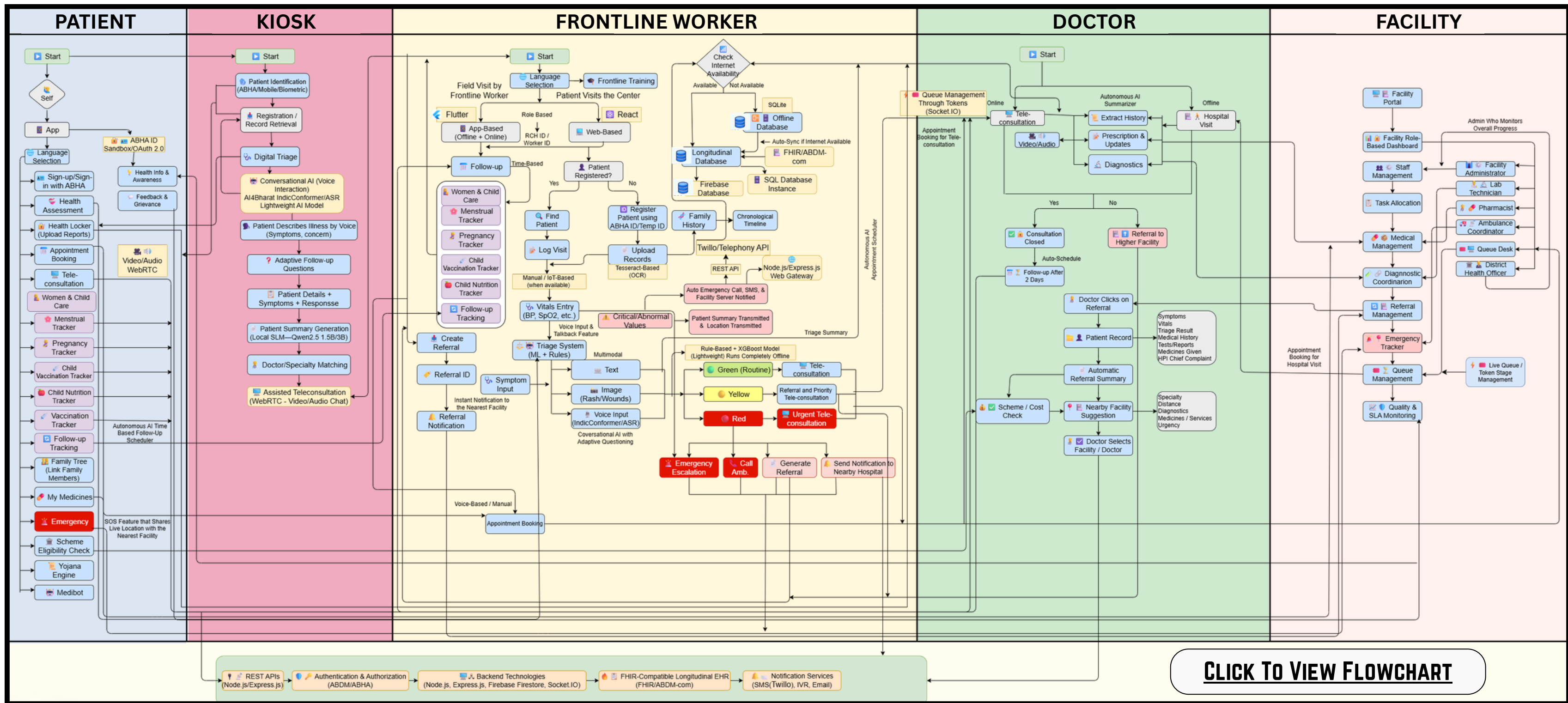
One Patient Journey, Multiple Access Point

Mobile App, Assisted Health Kiosks and Web Dashboards work on a shared backend for coordinated, end-to-end care.

Offline First | Voice Enabled

Core workflows run offline using lightweight models, while voice and conversational AI enable easy access to care with local data storage and automatic synchronization.

TECHNICAL APPROACH



WEB FRONTEND



APP FRONTEND



BACKEND



AI/ML



AI COMPONENTS



ML-Powered Triage — AI-driven triage using a rule-based decision tree, followed by ML-based final analysis.

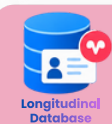


Smart Scheduling & Follow-Up — Voice interaction, automated appointment booking, and automated follow-up scheduling after two days.



Data Extraction — Tesseract-based OCR for automated extraction and processing of medical data.

DATA FALLBACK & SYNCHRONIZATION



Longitudinal Database — Centralized database maintaining patient health data across the entire portal, enabling a unified patient record throughout the system.



Offline Data Storage — SQLite-based local storage securely stores data when internet connectivity is unavailable, ensuring uninterrupted system functionality.



Auto Synchronization — Automatically syncs locally stored data with the central longitudinal database once connectivity is restored, keeping patient information consistent and up to date.

PROTOTYPE



SOURCE CODE



WEB DASHBOARD



MOBILE APP

WEB DASHBOARD

MOBILE APP

FEASIBILITY AND VIABILITY

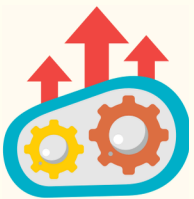
FEASIBILITY

Technical Feasibility



Uses lightweight AI models capable of offline/on-device processing, enabling triage and essential healthcare **services even in low-connectivity** areas. Voice-based and multilingual interaction further makes the system practical and **accessible for rural users**.

Operational Feasibility



SwasthyaConnect provides a simple, voice-first interface for frontline workers, making field-level use easier with **minimal training**. Its FHIR- and ABDM-compatible architecture supports **standardized patient records** and integration with existing digital healthcare workflows.

Economic Feasibility



Healthcare facilities benefit from better staff and workload management, smoother telemedicine operations, coordinated diagnostics and medicine availability, faster referrals, and real-time queue and emergency management. This **improves overall facility efficiency and resource utilization**.

GIGW 3.0 ALIGNED

Accessible | Usable
Government-Focused UI/UX

ADOPTION VIABILITY

Voice, multilingual interaction and simple interfaces reduce digital barriers for patients and frontline workers.

SCALABILITY VIABILITY

Modular App, Kiosk and Web architecture can expand across facilities, users and regions.

CHALLENGES

Integrating with existing rural healthcare workflows and ensuring user adoption



Language & Accessibility

Multiple regional languages and varying digital literacy across rural users.



Workflow Integration

Connecting patients, frontline worker doctors and facilities without disrupting existing workflows.



Kiosk Usability

Simple touch and voice interaction for assisted community access.



Hands-on Adoption

Minimal training and intuitive role-based interfaces for field workers and providers.

Ensuring reliable triage and care coordination in low connectivity areas



Offline AI

Lightweight models must deliver useful triage even without reliable internet.



IoT / Vital Variability

Different devices and noisy readings require validation and consistency checks.



Referral Accuracy

Correct routing depends on patient urgency, required services and facility availability.



AI Validation

Triage recommendations require safeguards, explain ability and clinician oversight.

Managing health data securely across offline & online channels



Auto Synchronization

Reliable sync must handle delayed uploads and conflicting updates.



Privacy & Security

Protect sensitive health information during storage, transmission and sharing.



Consent & Access Control

Role-based, consent-driven access across patients, workers, doctors and facilities.



Offline Data Storage

Patient data must remain available locally during connectivity loss.

SUSTAINABILITY VIABILITY

A shared backend enables phased deployment and incremental scaling without separate systems for each stakeholder.

IMPACTS AND BENEFITS

Economic Benefits

- **Lower Patient Costs**– Reduces expenses from long-distance travel, repeated visits and delayed referrals.
- **Timely & Affordable Care** – Teleconsultation improves access while reducing unnecessary healthcare costs.
- **Optimized Resources** – Digital workflows improve utilization of staff, medicines and diagnostics.
- **Efficient Facilities** – Better coordination increases facility and workforce efficiency.

Social Benefits

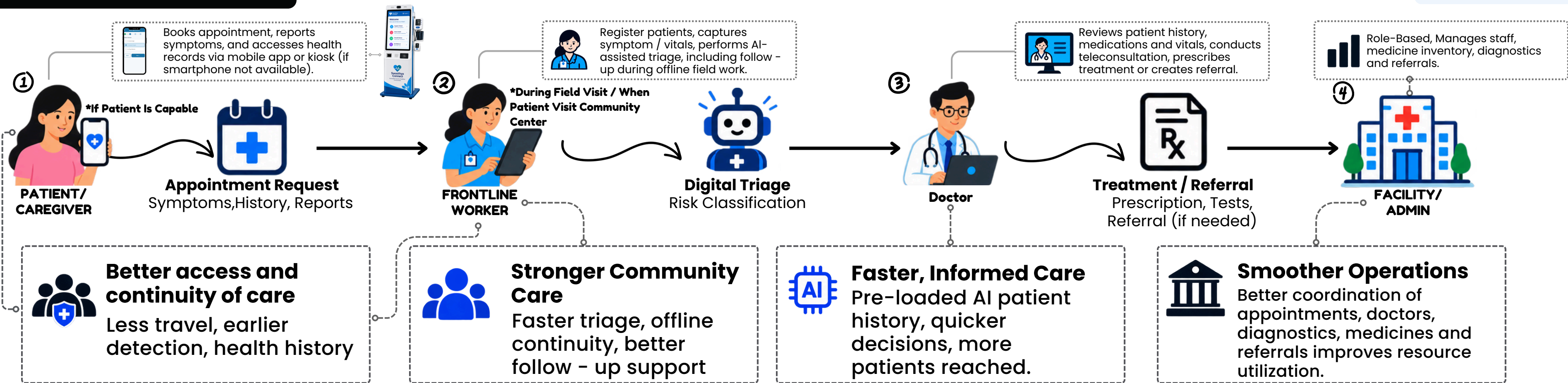
- **Improved Rural Access** – Reduces travel and waiting time through multilingual, kiosk-based care.
- **Stronger Frontline Care** – Increases frontline worker productivity and enables continuous patient follow-up.
- **Inclusive & Continuous Care** – Supports maternal, child and chronic care, including access to female doctors.
- **Greater Health Awareness & Safety** – Improves scheme awareness through MediBot & Yojna Engine and enables faster emergency response.

Environmental Benefits

- **Lower Travel Emissions** – Teleconsultation, smart referrals and remote follow-ups reduce unnecessary travel.
- **Reduced Paper Usage** – Digital prescriptions, OCR-based records and e-referrals minimize paperwork.
- **Efficient Resource Use** – Digital coordination reduces redundant visits and improves healthcare resource utilization.
- **Sustainable Healthcare Delivery** – Longitudinal EHRs enable paperless, digitally coordinated care.

STAKEHOLDERS & IMPACTS

Sample Scenario



OUR PROMISE

Our solution perfectly aligns with the UN Sustainable Development Goals.



How Swasthya_Connect Contributes

Government (Policy Planning):

Enables data-driven public-health planning, targeted resource allocation, better workforce and facility utilization, identification of underserved communities, and improved awareness and utilization of eligible government health schemes through the Yojna Engine.

Swasthya Connect is designed to **reduce care delays** in rural healthcare through offline-first AI, assisted kiosks, smart referrals and automated follow-up.



EARLIER RISK IDENTIFICATION

Timely detection of high-risk cases.



FASTER REFERRAL & ESCALATION

High-risk patients reach the right care faster.



BETTER FOLLOW-UP CONTINUITY

Ongoing tracking across facilities and care providers.



CONNECTED CARE FOR RURAL COMMUNITIES

Improved access, outcomes and equity.

India's Digital Health & Telemedicine Framework

Telemedicine Practice Guidelines define key requirements for remote consultations, including patient/RMP identification, consent, communication, evaluation, and patient management.

[CLICK HERE](#)

DPDP Act Compliance:

Consent-based data sharing, role-based access, secure processing and user control over personal health data.

[CLICK HERE](#)

ABDM establishes interoperable digital-health standards and supports secure, consent-based exchange of health information across healthcare stakeholders

[CLICK HERE](#) [CLICK HERE](#)

COMPARISON TABLE

Aspect	Existing Government Platforms (eSanjeevani, ABDM/ABHA, ANMOL, RCH, e-Aushadhi, etc.)	SwasthyaConnect Unified • Offline-first • Continuous Care
 Offline Access	Mostly online platforms with limited offline support.	 Offline-first mobile + kiosk Works without internet, encrypted auto-sync when connected
 Security & Privacy	ABDM / ABHA provides digital health records and interoperability	 Role-based access & data encryption. Patient consent-based sharing
 Continuity of Care	Records maintained across separate programs (ABDM, RCH, ANMOL, etc.).	 Unified longitudinal health record. Covers maternal, child & chronic care with. AI summaries, diagnostics, medicines & teleconsultation.
 Usability	Platform-specific interfaces for citizens and healthcare workers.	 Voice-enabled, multilingual & simple UI. For patients, frontline workers, doctors and facilities.
 Integrated Healthcare	Multiple independent platforms (eSanjeevani, ANMOL, RCH, e-Aushadhi, etc.).	 One unified platform. Teleconsultation, triage, referrals, diagnostics and follow-up
 Risk Detection	Program-specific tracking (e.g., RCH for maternal & child health).	 AI-powered triage & risk detection. Using symptoms, vitals and trends
 Medicine Management	e-Aushadhi for medicine supply and availability	 Real-time inventory view + e-prescription. With availability tracking
 Referral Management	Referral workflows vary by platform with limited end-to-end tracking.	 Unique referral ID with real-time tracking. Notifications and automatic follow-up
 Emergency Support	Limited emergency support and manual escalation.	 Automated escalation to nearest facility. Alerts to frontline worker, doctor and ambulance

DEMO VIDEOS

[LINK](#) [CLICK TO VISIT LINKS](#)

DOCUMENTATION

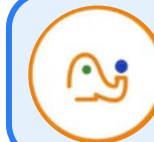
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ALL RESOURCES

[LINK](#)

GIGW 3.0

Government-oriented accessibility, usability & user-centric UI/UX

[CLICK HERE](#)

AI4BHARAT

Indic-language speech recognition multilingual voice interaction

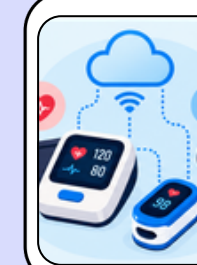
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REFERENCES



KIOSK

Explores an AI-assisted telemedicine kiosk combining health assessment, diagnostic inputs and remote doctor consultation.

[CLICK HERE](#)

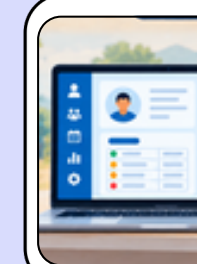
IOT

Develops an IoT system that captures parameters through sensors and transmits them for real-time remote health monitoring.

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HEALTH APP

Research shows mobile health tools can support frontline workers with digital care workflows, decision support and patient tracking in rural India.

[CLICK HERE](#)

WEB SYSTEM

Research from a rural PHC in Haryana examines web-based health portals for capturing, managing and using healthcare information.

[CLICK HERE](#)

EXISTING SOLUTIONS

[E-SANJEEVNI](#)[ABDM](#)[RCH](#)[ANMOL](#)[JANAUSHADHI](#)[NATIONAL NCD](#)